EE4002/EE4061: Introduction to Image Processing

HW3

Deadline: May 23, 2025

Question 1 (50 Points)

A discrete-time causal LTI system has the system function

$$H(z) = \frac{(1 + 0.2z^{-1})(1 - 9z^{-1})}{(1 + 0.81z^{-2})} \tag{1}$$

1. Is the system stable?
2. Determine expressions for a minimum-phase system $H_1(z)$ and an all-pass system $H_{ap}(z)$ such that

$$H(z) = H_1 z H_{ap}(z) \tag{2}$$

Question 2 (50 Points)

Many properties of a discrete-time sequence h[n] or an LTI system with impulse response h[n] can be discerned from a polezero plot of H(z). In this problem, we are concerned only with causal systems. Clearly describe the z-plane characteristic that corresponds to each of the following properties:

1. Real-valued impulse response
2. Finite impulse response
3. $h[n] = h[2\alpha - n]$ where 2α is an integer
4. Minimum phase
5. All-pass